

# Equivalence Completion in the QCD CP–Odd Sector

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## Abstract

We complete an equivalence map in the QCD CP–odd sector forced by admissibility, standing conservation, and closure. Parameters commonly treated as independent in ultraviolet gauge bookkeeping are shown to overcount tensor load. Admissible redescription and ultraviolet-to-infrared transport reduce the CP–odd content to a single invariant. The result is eliminative, invariant under admissible redescription, and introduces no new primitives or physics.

## 1 Apparent Independence in Ultraviolet Bookkeeping

In ultraviolet gauge bookkeeping, CP–odd structure appears in two operator classes:

$$\mathcal{L}_\theta = \theta \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad (1)$$

$$\mathcal{L}_m = -\bar{q}_R M q_L - \bar{q}_L M^\dagger q_R, \quad (2)$$

where  $M$  is a complex quark mass matrix. The parameter  $\theta$  and the phases of  $M$  are commonly treated as independent inputs in this skin.

## 2 Admissible Redescription

Consider a global chiral rephasing of quark fields,

$$q \rightarrow e^{i\alpha\gamma_5} q. \quad (3)$$

Under this admissible redescription:

- The phase of  $\det M$  shifts by  $-2N_f\alpha$ .
- The coefficient  $\theta$  shifts by  $+2N_f\alpha$ .

Thus neither  $\theta$  nor  $\arg \det M$  is invariant under admissible redescription. Only specific combinations may survive as invariant tensor content.

## 3 Ultraviolet–to–Infrared Transport

An admissible low-energy (chiral/EFT) description of the same hadronic CP–odd physics exposes a single CP–violating parameter governing all CP–odd hadronic observables. No admissible infrared skin provides independent invariant witnesses corresponding separately to  $\theta$  and  $\arg \det M$ .

## 4 Lemma: Invariant Witness Counting

Let a tensor load possess  $n$  independent degrees of freedom. Under admissible redescription and regime transport, each degree must correspond to an invariant witness in every admissible skin describing the same regime. If an admissible skin exposes fewer invariant witnesses, the higher-dimensional parameterization necessarily contains redundancy.

## 5 Failure of Independence

Assume  $\theta$  and  $\arg \det M$  are independent tensor degrees of freedom. By invariant witness counting, two invariant witnesses must survive admissible redescription and ultraviolet-to-infrared transport. Only one invariant witness exists. Preserving independence would require loss of tensor load, hidden unwitnessed structure, or representational privilege. Each constitutes standing failure.

## 6 Forced Tensor Relation

The admissible CP-odd tensor load is uniquely specified by the invariant combination

$$\boxed{\bar{\theta} \equiv \theta + \arg \det M} \tag{4}$$

(up to purely conventional sign choices). Neither  $\theta$  nor  $\arg \det M$  is independently admissible as tensor content.

## 7 Corollaries

- The QCD CP-odd sector admits a one-dimensional tensor load.
- All admissible ultraviolet and infrared descriptions of strong CP violation are functions of  $\bar{\theta}$  alone.
- Any framework treating  $\theta$  and mass phases as independently physical overcounts tensor degrees and violates standing.

## Conclusion

Equivalence completion enforces collapse of apparent freedom in the QCD CP-odd sector. The result is eliminative, representation-invariant, and irreversible under admissible redescription.